

Repsol Bio Electra

Vegetable ester dielectric oil — Uninhibited

PRODUCT DATA SHEET

Description

BIO ELECTRA dielectric oil is formulated with over 99% vegetable oils and contains no synthetic antioxidants. Its composition does not include silicones, halogens, or any other component that may pose a risk to health or the environment. Thanks to its high vegetable oil content, it is rapidly biodegradable and non-toxic for aquatic and land-based ecosystems. This oil is very safe thanks to its high ignition point, reducing the risk of fire and its consequences. It is recommended for use in new electrical equipment where there is no contact with air.

Classification and standards

BIO ELECTRA has an ignition point of over 300 °C and is classified as a type K fluid according to IEC 61100. It complies with the IEC 62770: 2014 standard.

Technical specifications

Values according to IEC 62770: 2013 standard.

Properties	Units	Method	Min.	Max.	Typical value
Physical properties					
Appearance		ASTM D1524		Bright and clear	
Density at 20 °C	g/cm ³	ASTM D4052		1.00	0.91
Viscosity at 0 °C	mm ² /s	ASTM D445		500	275.9
Viscosity at 40 °C	mm ² /s	ASTM D445		50	39.2
Viscosity at 100 °C	mm ² /s	ASTM D445		15	8.5
Pour point	°C	ASTM D97		-10	-25
Coefficient of thermal expansion 0-50 °C	°C ⁻¹	ASTM D1903			0.00072
Thermal conductivity at 25 °C	W/K m	ASTM D2717			0.1691
Specific heat at 25 °C	J/K g	ASTM D2766			1.97
Chemical properties					
Soluble acidity	mg KOH/g	IEC 62021-3		0.06	0.05
Water content	mg/kg	IEC 60814		200	150
DBDS	mg/kg	IEC 62697-1		Undetectable	
Electrical properties					
Dielectric dissipation factor at 90 °C, 50 Hz		IEC 60247		0.05	0.03
Breakdown voltage (on delivery)	kV	IEC 60156	35		65
Electrical conductivity at 25 °C	pS/m	ASTM D2624			3
Dielectric constant at 25 °C		IEC 60247			3.1
Gassing tendency	µl/min	IEC 60628A			-31.2
Oxidation stability — IEC 61125C					

Properties	Units	Method	Min.	Max.	Typical value
Total acidity	mg KOH/g	IEC 61125C		0.6	0.38
Viscosity at 40 °C (% increase)	%	ISO 3104		30	14.1
Dielectric dissipation factor at 90 °C, 50 Hz		IEC 60247		0.500	0.120
Health, safety and the environment					
Fire point	°C	ASTM D92	300		362
Flash point	°C	ASTM D92	250		330
Biodegradability after 28 days	%	OECD 301B	60		85
Aquatic ecotoxicity	mg/l	OECD 201/202/203	100		>1000
Terrestrial ecotoxicity	mg/kg	OECD 207/208	100		>1000

Note: Typical values correspond to current production. Whilst future production will conform to Repsol's specification, variations in these characteristics may occur. The guaranteed minimum/maximum values define the product specification.